

CODE - SYNCRONIX

A Major Project Report Submitted to



**Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal
Towards partial fulfillment for the award of**

**Bachelor of Technology
Computer Science & Engineering**

Submitted by

- 1. Kuldeep singh Rathore(0818CS2011079)**
- 2. Naman Jain (0818CS201103)**
- 3. Pradyumn Tahekar(0818CS201122)**

Under the supervision of

**Ms. Reetu Gupta
Assistant Professor**



**Department of Computer Science & Engineering
Indore Institute of Science and Technology, Indore
Session 2023-2024**

Indore Institute of Science and Technology, Indore (M.P.)

Department of Computer Science & Engineering

DECLARATION

We, **Kuldeep Singh Rathore** (0818CS2011079) , **Pradyumn Tahekar** (0818CS201122) **Naman Jain** (0818CS201103) students of **Bachelor of Technology in Computer Science & Engineering** discipline, **Indore Institute of Science and Technology, Indore (M.P.)**, hereby declare that the work presented in this project entitled “**Code-Synchronix**” is the outcome of our own work, is bonafide and correct to the best of my knowledge and this work has been carried out taking care of Engineering Ethics. The work presented here does not infringe any patented work and has not been submitted to any other university or anywhere else for the award of any degree or any professional diploma.

1.Kuldeep Singh Rathore (0818CS2011079)

2. Naman Jain (0818CS201103)

3 Pradyumn Tahekar (0818CS201103)

Date: 01/12/2023

Indore Institute of Science and Technology, Indore (M.P.)

Department of Computer Science & Engineering

RECOMMENDATION

This is to certify that the work embodied in this project work entitled “**Code-Syncronix**”, being submitted by **Kuldeep Singh Rathore** (0818CS2011079) , **Naman Jain**(0818CS201103) **Pradyumn Tahekar** (0818CS201122)in partial fulfillment of the requirement for the award of **Bachelor of Technology in Computer Science & Engineering** discipline, to **Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal (M.P.)** during the academic year 2023-24 is a record of bonafide piece of work, carried out by her under my supervision and guidance.

Mrs. Reetu Gupta

Assistant Professor)

Project Guide

Dr.Richa Gupta

HOD CSE

Forwarded by:

Dr. Keshav Patidar

Principal, IIST, Indore

Indore Institute of Science and Technology, Indore (M.P.)

Department of Computer Science & Engineering

CERTIFICATE

The Project entitled “**Code-Syncronix**” being submitted by **Kuldeep Singh Rathore** (0818CS2011079), **Naman Jain** (0818CS201103) and **Pradyumn Tahekar** (0818CS201122) has been examined by us and is hereby approved for the award of degree **Bachelor of Technology** in **Computer Science & Engineering** discipline, for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but approve the dissertation only for the purpose for which it has been submitted.

(Internal Examiner)

Date:

(External Examiner)

Date:

ACKNOWLEDGEMENT

After the completion of this project, words are not enough to express my feelings about all those who helped me to reach my goal; feeling above this is my indebtedness to the Almighty for providing me this moment in life.

In this project we have received constant support from Dean Academics & Head of Department **Dr. Richa Gupta**. Also, I am heartily indebted to the constant support and guidance of **Mrs. Reetu Gupta**. Without her guidance and scholarly suggestion an urge to bring out the best would not have been possible. I hope to propagate his scientific, industrial, and professional fervors to the best of my abilities. His/her clear view and knowledge provided help during every phase of Project Development. His perpetual motivation, patience, and excellent expertise in discussion during progress of the project work have benefited us to an extent, which is beyond expression. His/her in-depth and breadth of knowledge of Computer Engineering field made me realize that theoretical knowledge always helps to develop efficient operational software, which is a blend of all core subjects of the field. She was major support to me throughout my project, being available with his/her odd ideas, inspiration, and encouragement. It is through his masterful guidance that I have been able to complete my Project.

I am also thankful to all the teaching and non-teaching staff from Department of Computer Science and Engineering and friends and people who helped me directly or indirectly for the completion of this project, with success.

The successful completion of a project is generally not an individual effort. It is an outcome of the cumulative effort of many people, each having their own importance to the objective. This section is a vote of thanks and gratitude towards all those persons who have directly or indirectly contributed in their own special way towards the completion of this project.

Date:

1. Kuldeep Singh Rathore(0818CS2011079)

2. Naman Jain (0818CS201103)

3. Pradyumn Tahekar(0818CS201122)

ABSTRACT

"Code Synchronix" is a cutting-edge real-time web application revolutionizing collaborative coding experiences. At its core, the platform facilitates the seamless collaboration of multiple users in virtual coding rooms, where they can collectively write, debug, and compile code in real-time synchronization. The primary goal is to eliminate geographical barriers and enhance the efficiency of group coding projects by providing a dynamic and interactive environment. By connecting users to shared coding spaces, the application ensures that every change made by one participant is instantly reflected for all others, fostering a truly synchronized and immersive coding experience.

One of the key features of Code Synchronix is its support for multi-user editing, allowing simultaneous contributions to the codebase. Each user's input is color-coded, providing a clear visual representation of individual contributions and enhancing overall visibility within the collaborative environment. Live debugging is another standout feature, empowering users to collectively identify and rectify issues as they arise. The platform's integrated compiler and execution environment further streamline the development process, offering instant feedback on code functionality within the collaborative space.

In addition to its technical capabilities, Code Synchronix incorporates version control features, allowing users to track changes, revert to previous states, and maintain a comprehensive history of the collaborative coding session. The user-friendly interface, complete with chat functionality, code commenting, and real-time notifications, ensures accessibility for developers of all skill levels. Security and privacy are paramount, with the platform implementing secure authentication mechanisms and encrypted communication channels to create a safe and confidential coding environment.

In essence, Code Synchronix stands as a testament to the future of collaborative coding. Whether used for educational purposes, remote team collaboration, or spontaneous coding sessions, this web application offers a versatile and efficient space where developers can collectively bring their ideas to life in real-time synchronization. By combining advanced technical features with a user-friendly interface, Code Synchronix represents a new frontier in collaborative coding experiences.

TABLE OF CONTENTS

TITLE	PAGE
NO.	
Declaration	2
Recommendation	3
Certificate	4
Acknowledgement	5
Abstract	6
Table of Contents	7
List of Figures	10
1. INTRODUCTION	11
1.1 Introduction	11
1.2 Objective	13
2. LITERATURE SURVEY	14
2.1 Background and Related Work	14
2.2 Problem Statement	14
2.3 Objective	15
3. PROPOSED METHODOLOGY	16

3.1 Proposed Method	14
3.2 Proposed Methodology Architecture	15
3.3 Proposed Methodolgy Flow Chart	16
3.4 System interface	17
4. SYSTEM REQUIREMENT SPECIFICATIONS	19
4.1 Hardware Requirements	19
4.2 Software Requirements	19
4.3 Tools/Libraries/APIs used	19
5. SYSTEM ANALYSIS, DESIGN, AND IMPLEMENTATION	20
5.1 System Analysis	20
5.1.1. Feasibility Analysis	20
5.2 System Design	21
5.2.1. Architecture/Component Diagram	21
5.2.2. Data Flow Diagrams (DFD)	21
5.2.3. Use-Case Diagram	22
5.2.4. Sequence Diagram	22
5.2.5. Activity diagram	23
5.3 Implementation	24
5.4 Snapshots of our Project	25
6. RESULTS	27
7. CONCLUSION AND FUTURE WORK	28
7.1 Conclusion	28
REFERENCES	29
APPENDIX	

LIST OF FIGURES

FIGURE NO.

PAGE NO.

3.2. Flowchart of proposed system

5.2.1. Architecture/Component Diagram

5.2.2. Data Flow Diagrams (DFD)

5.2.3. Use-Case Diagram

5.2.4. Sequence Diagram

5.2.5. Activity Diagram

5.3.1. Snapshots of our Project

6. Results

INTRODUCTION

"Code Synchronix" represents a groundbreaking leap in the realm of collaborative coding, offering an innovative solution for real-time coding experiences. In a digital landscape where distance can often impede seamless teamwork, this web application aims to transcend geographical boundaries by creating virtual coding rooms. These dynamic spaces empower multiple users to work collaboratively on coding projects, allowing them to write, debug, and compile code synchronously. By facilitating immediate and shared access to the coding environment, Code Synchronix fosters a sense of real-time connection among developers, irrespective of their physical locations.

The heart of this project lies in its commitment to providing a comprehensive and interactive platform for users of varying expertise. The application not only supports multi-user editing, enabling simultaneous contributions, but also integrates live debugging features. Developers can collectively identify and address issues in real time, enhancing the efficiency of collaborative coding sessions. Moreover, the platform includes a built-in compiler and execution environment, offering instant feedback on the code's functionality. As a result, Code Synchronix emerges as a user-friendly, secure, and powerful tool that redefines the collaborative coding landscape, promising an immersive and synchronized experience for developers worldwide.

At its core, Code Synchronix addresses the evolving needs of developers and coding teams by providing a collaborative environment that transcends traditional constraints. The platform is designed to be adaptable to various scenarios, whether it's facilitating remote teamwork, supporting collaborative coding education, or enabling spontaneous coding sessions among enthusiasts. By leveraging cutting-edge technologies, Code Synchronix aims to enhance the productivity and creativity of developers by fostering a shared space where ideas can be exchanged, tested, and refined in real time.

The user experience is a central focus of Code Synchronix, featuring an intuitive interface that promotes ease of use. Through color-coded contributions, the platform ensures transparency and clarity, allowing users to easily identify and track individual changes within the collaborative workspace. The inclusion of chat functionality and real-time notifications further augments communication, promoting a sense of community and collaboration among users.

1.2. Objective

The Code Synchronix project aims to redefine collaborative coding by addressing key objectives that prioritize efficiency, accessibility, and innovation. Through the creation of virtual coding rooms, the project seeks to break down geographical barriers and enable seamless collaboration among multiple users in real-time. By fostering a dynamic and interactive coding environment, the project's primary goal is to enhance the productivity and creativity of developers, offering a platform where they can collectively write, debug, and compile code without the constraints of physical proximity.

Code Synchronix focuses on supporting multi-user editing, allowing simultaneous contributions with color-coded differentiation for clear visibility. Live debugging features empower developers to collaboratively identify and address issues, promoting quicker issue resolution and an iterative development process. The integrated compiler and execution environment provide instant feedback on code functionality, streamlining the coding and testing processes within the collaborative space.

The project prioritizes user-friendly design, incorporating an intuitive interface with chat functionality, code commenting, and real-time notifications. Furthermore, Code Synchronix places a strong emphasis on security and privacy, implementing robust measures such as secure authentication and encrypted communication channels to safeguard users' coding sessions.

Versatility is a key aspect of Code Synchronix, as it aims to cater to various scenarios, including remote team collaboration, collaborative coding education, and spontaneous coding sessions. By embracing emerging technologies and envisioning the future of collaborative coding, Code Synchronix positions itself as a forward-looking project that anticipates the evolving needs of the coding community, offering an immersive, synchronized, and efficient coding experience for developers worldwide.

2. LITERATURE SURVEY

2.1. Background and Related Work

In the ever-evolving landscape of software development, the collaborative nature of coding has become increasingly vital. Traditional development environments often face challenges when it comes to real-time collaboration, especially in scenarios where team members are geographically dispersed. The Code Synchronix project emerges from the recognition of this need for a dynamic, virtual space where developers can collaborate seamlessly in real-time. With the aim of overcoming geographical barriers and enhancing the efficiency of group coding projects, Code Synchronix seeks to redefine the collaborative coding experience by providing an interactive platform that promotes simultaneous coding, debugging, and compilation.

The project draws inspiration from the growing demand for remote work solutions, the rise of collaborative coding practices, and the need for versatile tools that can adapt to diverse coding scenarios. The objective is not only to address the technical challenges of real-time synchronization but also to create an environment that fosters creativity, communication, and efficient problem-solving among developers.

Several existing tools and platforms have paved the way for collaborative coding, contributing valuable insights and solutions to similar challenges. Platforms like Visual Studio Live Share, Google Docs for Code, and CodePen have explored aspects of real-time collaboration. Visual Studio Live Share, for instance, allows developers to collaboratively edit and debug code in real time. Google Docs for Code focuses on collaborative editing, and CodePen provides an online environment for front-end development collaboration.

While these tools offer valuable features, Code Synchronix seeks to differentiate itself by providing a comprehensive and versatile solution that combines real-time collaboration, live debugging, integrated compilation, and a user-friendly interface. By incorporating elements of version control, security measures, and adaptability to various coding scenarios, Code Synchronix aims to stand at the forefront of collaborative coding platforms, addressing the evolving needs of developers in a rapidly changing digital landscape.

2.2. Problem Statement

The Code Synchronix project is conceived against the backdrop of challenges prevalent in existing collaborative coding tools. Many tools in the literature provide real-time editing capabilities, yet fall short in delivering a holistic collaborative environment. In response, Code Synchronix not only enables simultaneous editing but also introduces features such as live debugging, integrated compilation, and synchronized code execution in real-time, offering a more comprehensive collaborative coding experience.

Geographical barriers often impede effective collaboration among dispersed teams, a limitation that Code Synchronix addresses through the creation of virtual coding rooms. By providing developers with a platform to seamlessly collaborate regardless of physical locations, the project aims to overcome these geographical constraints, fostering inclusivity and accessibility.

Security concerns in collaborative coding tools are a common issue, with potential threats to the confidentiality and integrity of coding sessions. Code Synchronix places a strong emphasis on security by implementing robust measures such as secure authentication mechanisms and encrypted communication channels, ensuring a safe and confidential coding environment for users.

Versatility is another challenge faced by some collaborative coding tools, often tailored to specific use cases. Code Synchronix aims to be adaptable to various scenarios, including remote team collaboration, collaborative coding education, and spontaneous coding sessions. This flexibility allows the platform to meet the diverse needs of the coding community.

Incomplete integration of features within existing tools can hinder the seamless flow of the coding workflow. Code Synchronix addresses this challenge by integrating real-time collaboration, live debugging, and integrated compilation, providing users with a cohesive and comprehensive coding experience.

Furthermore, Code Synchronix tackles the issue of complex user interfaces by prioritizing a user-friendly design. The platform incorporates intuitive features such as chat functionality, code commenting, and real-time notifications, enhancing communication and collaboration among users. In summary, Code Synchronix emerges as a solution to these challenges, offering a secure, versatile, and feature-rich collaborative coding platform that redefines the collaborative coding landscape for developers worldwide.

PROPOSED METHODOLOGY

3.1. Proposed Method

The development of Code Synchronix follows a structured methodology to ensure the successful implementation of key features and meet project objectives. Beginning with a thorough analysis of user and stakeholder requirements, the project identifies essential features such as real-time collaboration, live debugging, integrated compilation, security measures, and user-friendly interfaces.

A comprehensive system design is then developed, defining the architecture, data flow, and component interactions. Prototyping involves building an initial model to validate the design and gather user feedback, focusing on simultaneous code editing, live debugging, and integrated compilation. The subsequent development phase prioritizes security measures and version control features, adhering to best coding practices. Thorough testing, including usability testing, is conducted to identify and address any bugs or vulnerabilities.

Code Synchronix is then deployed in a controlled environment, and continuous deployment strategies are implemented for updates. Comprehensive documentation is created, including user guides and technical resources. Feedback and iteration are integral, with regular updates addressing user needs and technological advancements. The project emphasizes community building by engaging users, establishing forums, and considering open-sourcing aspects of Code Synchronix to encourage contributions and collaborations. This iterative process ensures flexibility in adapting to changing requirements and advancements in collaborative coding.

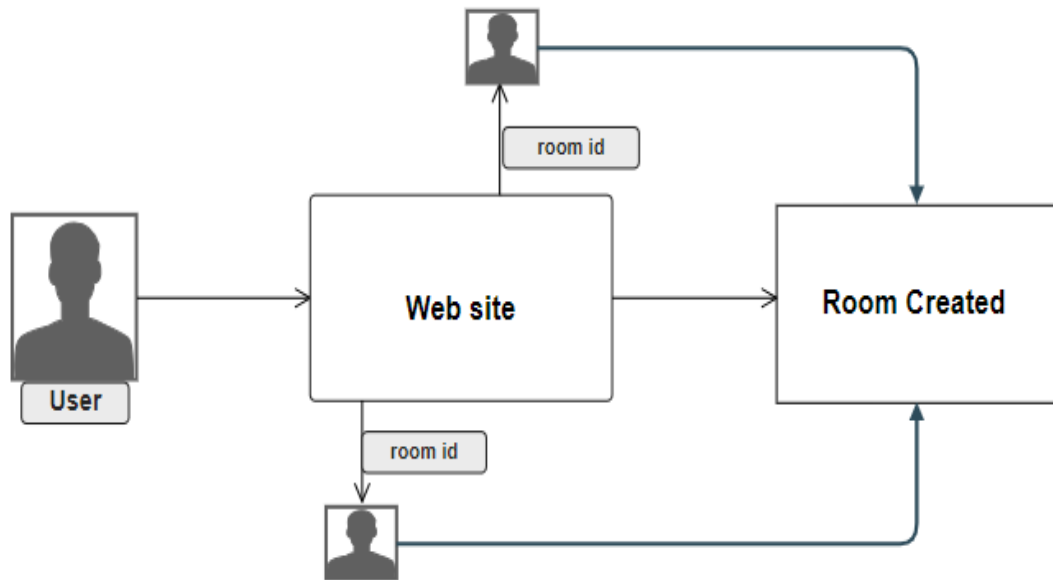
After deployment, Code Synchronix undergoes continuous monitoring to track its performance, user interactions, and potential issues. This ongoing monitoring helps identify areas for optimization and enhancements. Regular performance assessments ensure that the platform operates efficiently, meeting the evolving needs of users.

3.2. Proposed Methodology Architecture

Agile methodology is a set of principles and practices for software development that prioritizes flexibility, collaboration, and customer satisfaction. It emerged as a response to the challenges of traditional, linear development methodologies, emphasizing iterative and incremental development. The Agile Manifesto, created in 2001 by a group of software developers, outlines the core values and principles that guide Agile development.



3.3.Proposed Methodology Flow Chart



3.4.System Interface

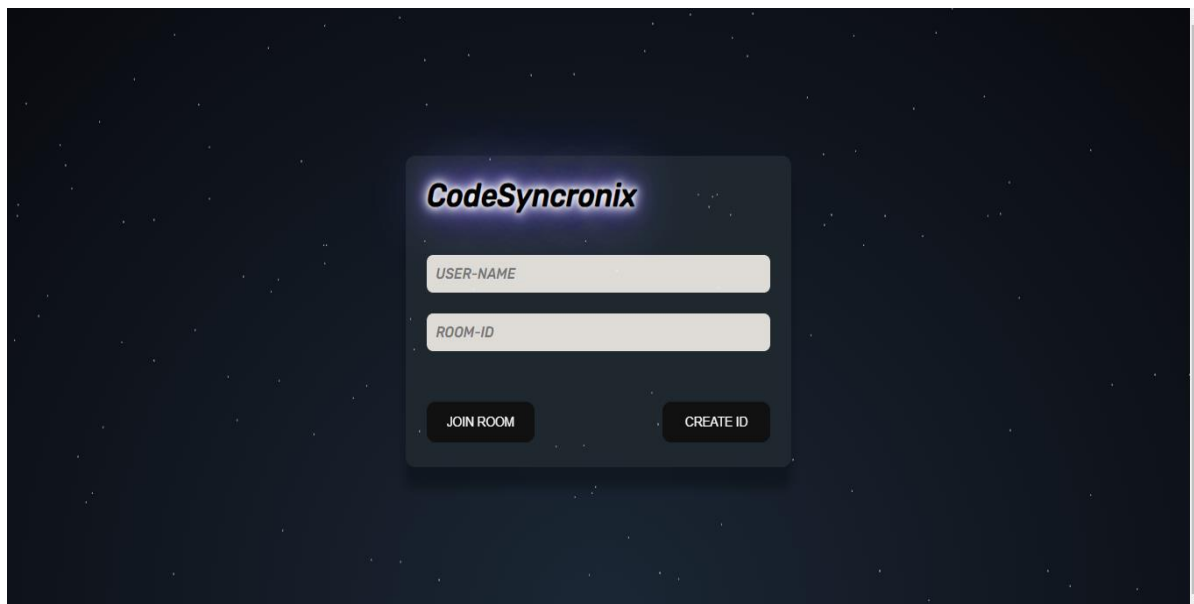


Figure 3.4.1 Main Page

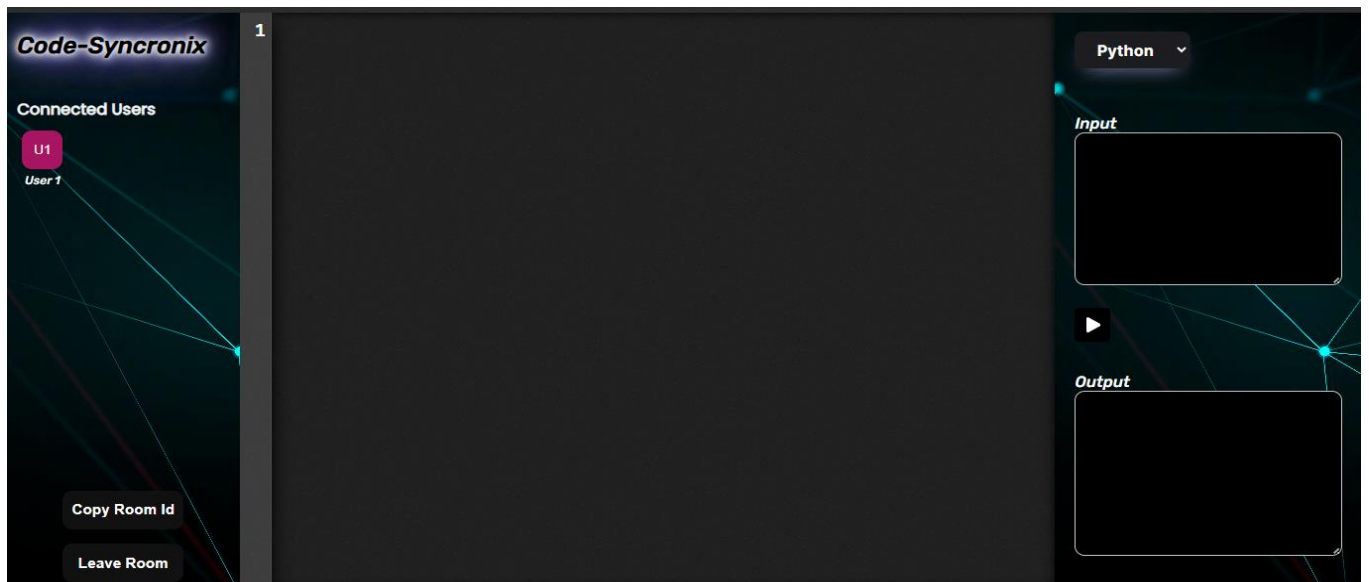


Figure 3.4.2 Room Created

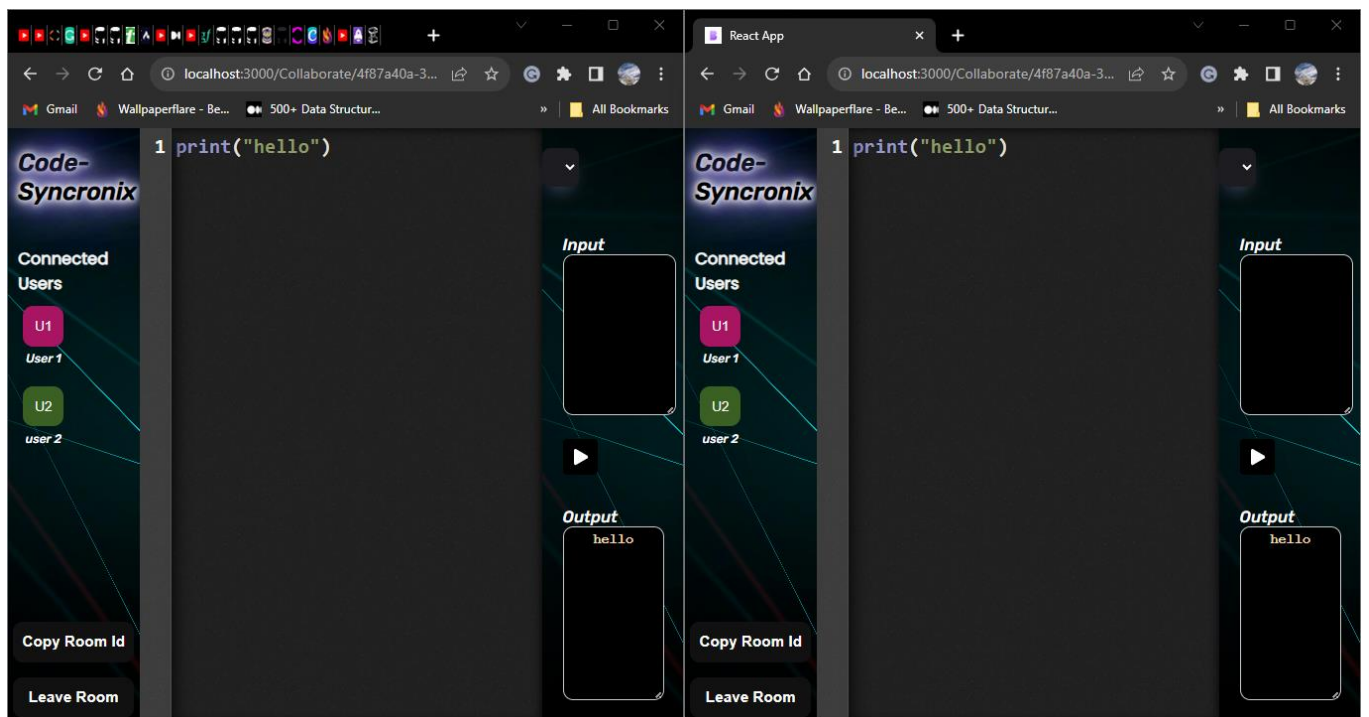


Figure 3.4.3 Two user Joined

SYSTEM REQUIREMENTS

The system requirement definition is concerned with the analysis of the existing system with the aim of determining and structuring the requirement of the proposed system. It is achieved with the aid of user requirement. The analysis stage was specifically carried out in focus of the functionality dataflow of project.

Platform

Windows is very powerful scalable Operating System that provides basic that provides basic file and prints services as well as robust platform for server application. Main features are as follows –

- An easier way to use Interface and tools.
- More extensive Network Performance.
- Enhanced communication features.

4.1. Hardware Requirements

- Processor: AMD Ryzen 5 4500U with Radeon Graphics 2.38 GHz
- RAM: 4.00 GB
- System type 64-bit operating system, x64-based processor
- Keyboard: Any
- Mouse: Any

4.2. Software Requirements For frontend:

- HTML,CSS/BOOTSTRAP,JAVASCRIPT
- VS-studio for project implementation.
- Operating System -Window 10 or higher.
- Web Browser -Mozilla/Google Chrome/Internet Explore

For backend:

- Node js

5. SYSTEM ANALYSIS, DESIGN AND IMPLEMENTATION

5.1. System Analysis

System Analysis is a detailed study of the various operations performed by a system and their relationships within and outside of the system. Here the key question is – what all problems exist in the present system? What must be done to solve the problem? Analysis begins when a user/student begins a study of the program using existing system. During Analysis, data collected on the various files, decision points and transactions handled by the present system. The success of the system depends largely on how clearly the problem is defined, thoroughly investigated and properly carried out through the choice of solution. A good analysis model should provide not only the mechanisms of problem understanding but also the frame work of the solution. Thus, it should be studied thoroughly by collecting data about the system.

5.1.1. Feasibility Analysis:

When a project is started an initial investigation is carried out. During these phase of study users need has recognized and other requirements are determined. Once the problem has been defined a study is carried out to select the best system i.e. a feasible system that meets performance requirements. So, feasibility is the determination of whether or not a project is worth doing and the process followed in making this determination is called a feasibility study. In order to conduct the feasibility study, we have seven distinct, but interrelated types of feasibility, these are technical feasibility, Operational feasibility, Economical feasibility, Social feasibility. Management feasibility, Legal feasibility, and Time feasibility.

These keys constraints are involved in feasibility study.

- Financial feasibility.
- Technical feasibility.

1. Financial and Economic Feasibility:

Economic feasibility or cost benefit is an assessment of the economic justification for a computer system project. Since the system is a web based, any number of users can use this tool from anywhere and anytime. Not only the cost of hardware, software etc, is considered but also benefits in the form of reduced cost are considered.

2. Technical Feasibility:

Technical feasibility is the assessment of the technical of the system. The develop for platform independent environment are used to develop the system. The technical feasibility has been carried out. The system is technical feasible for developed. The system can be developed with the existing facility. The technical consideration evaluates the hardware requirement, software technology etc...as per the requirement HTML, CSS as frontend,Node js as backend.

5.2. System Design

5.2.1. System Architecture diagram



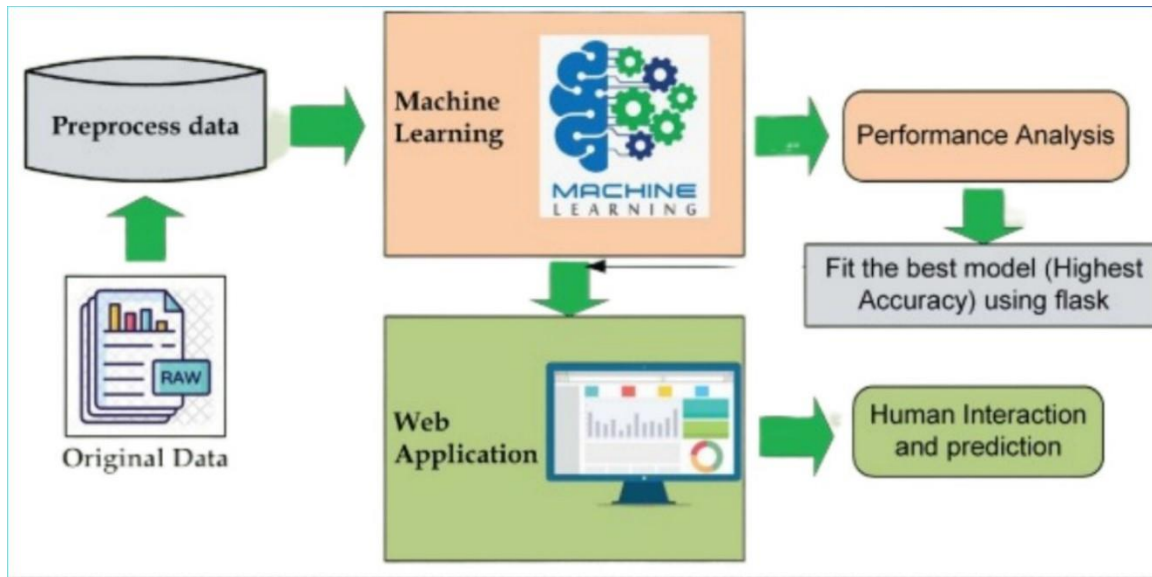
Code Synchronix is a real-time web application that allows multiple users to join a room and write, debug, and compile their code in real-time synchronization. The system architecture diagram above shows the different components of the system and how they interact with each other.

The Code Editor component is responsible for providing a user interface for writing and editing code. There can be multiple instances of the Code Editor component, one for each user in the room.

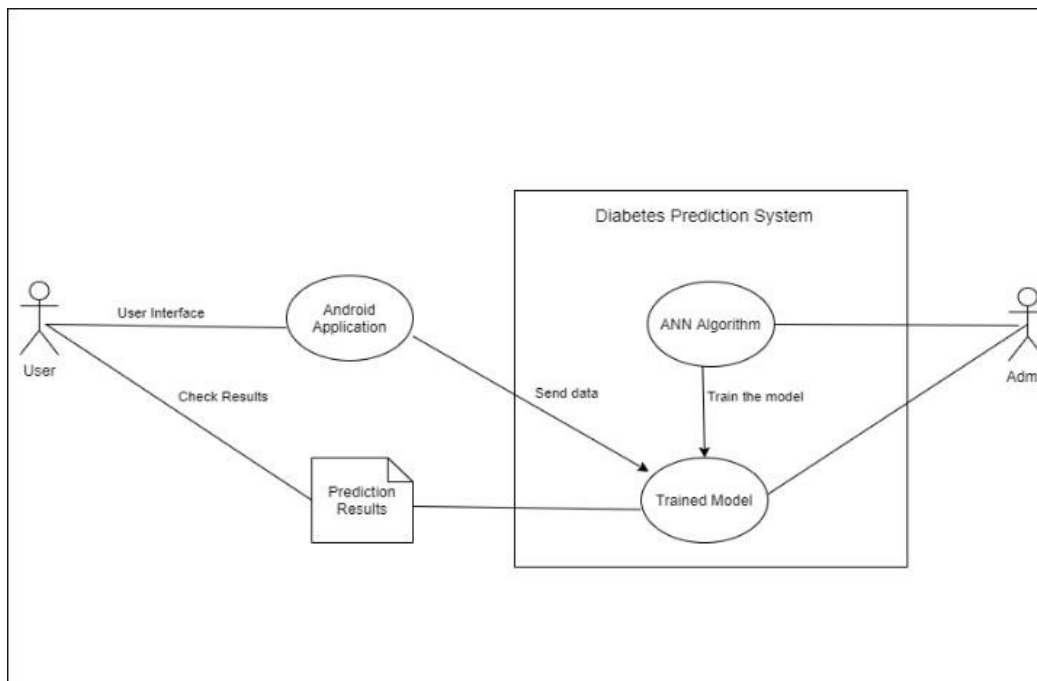
The Web Socket component is responsible for handling real-time communication between the Code Editor and the Server components. It allows users to see each other's changes in real-time.

The Server component is responsible for handling requests from the Code Editor and Web Socket components. It processes the code and sends the results back to the Code Editor component.

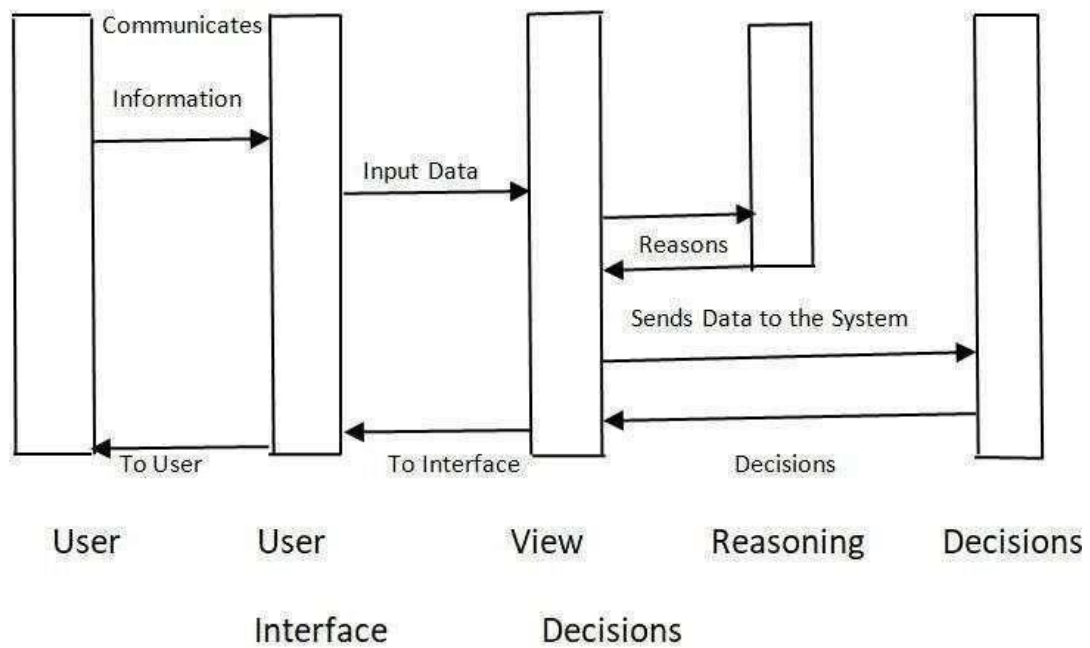
5.2.2. Data Flow Diagram (DFD)



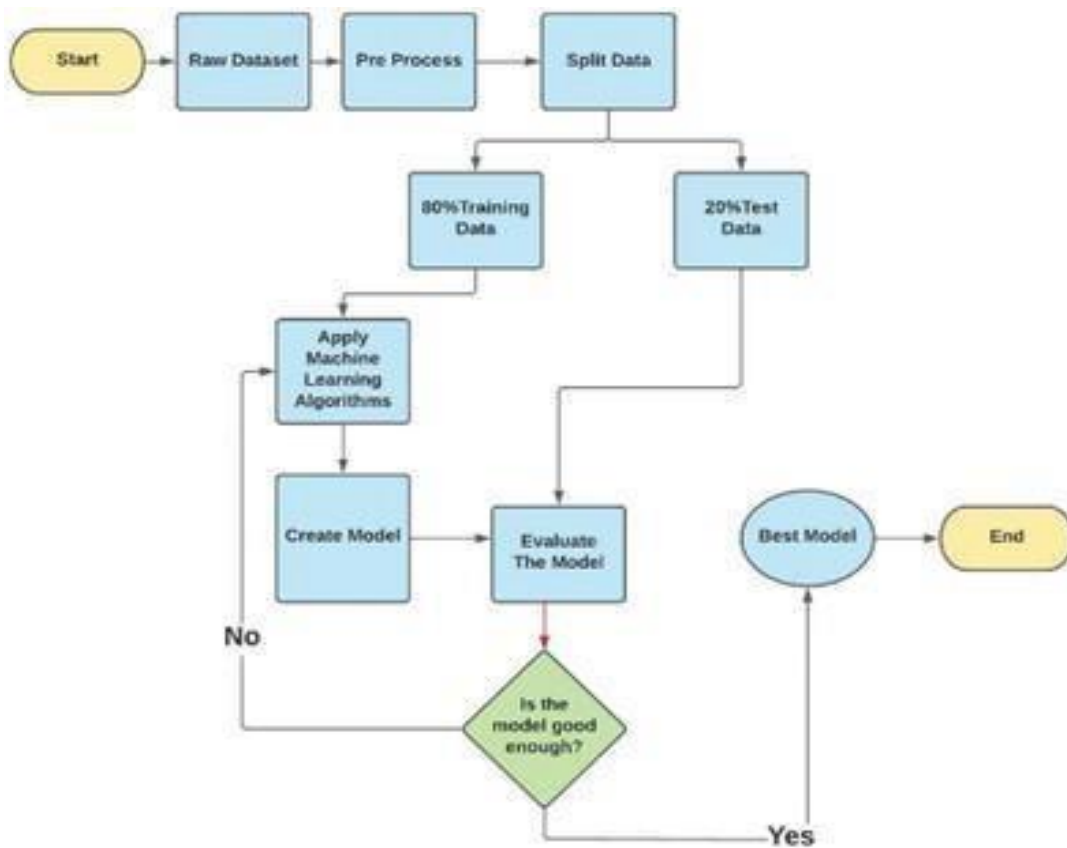
5.2.3. Use-Case Diagram



5.2.4. Sequence Diagram



5.2.5. Activity Diagram



5.3. Implementation

Implementation is the stage of the project when the theoretical design is turned out into a working system. Thus, it can be considered to be the most critical stage in achieving a successful new system and in giving the user, confidence that the new system will work and be effective. The implementation stage involves careful planning, investigation of the existing system and its constraints on implementation, designing of methods to achieve changeover and evaluation of changeover methods.

As we know, creating software is one thing and the implementation of the created software is another. The process of implementing software is much difficult as compared to the task of creating the project. First, we have to implement the software on a small scale for removing the bugs and other errors in the project and after removing them we can implement the software on a large scale.

Utilizing a dataset with features such as age, gender, BMI, and various diabetes symptoms, we meticulously analyzed and cleaned the information. Developing a robust machine learning model, we harnessed sophisticated algorithms to identify correlations related to diabetes risk factors. Alongside this model, we designed a user-friendly questionnaire addressing diabetes symptoms for user input. The model utilizes responses to predict an individual's likelihood of developing diabetes, presenting the result in a percentage format. This approach aims to enable early detection and proactive management, providing a promising tool for mitigating the risks associated with diabetes onset.

6. RESULTS

Our result will be in form of percentage score where the user will get to know how much percentage of diabetes risk they possess.

7. CONCLUSION

In conclusion, the development and implementation of machine learning algorithms for diabetes prediction have shown promising results in accurately assessing the risk factors and forecasting the likelihood of diabetes onset in individuals. Through the utilization of diverse datasets and advanced predictive models, this project has demonstrated the potential to aid in early detection and proactive management of diabetes.

Ultimately, the application of machine learning in diabetes prediction presents a promising avenue for augmenting preventive healthcare measures, enabling early intervention, and significantly reducing the burden of this chronic condition on individuals and healthcare systems. The ongoing advancements in this field hold tremendous potential to revolutionize diabetes care, paving the way for a future where proactive management and personalized interventions are key components in combating this prevalent health challenge.

REFERENCES

- Duca, Filippo A., et al. "Machine learning in the prediction of diabetes complications: a narrative review." *Journal of Diabetes Science and Technology* 13.4 (2019): 754-764.
- *Machine Learning in Medicine - A Complete Overview*, Ton J. Cleophas and Aeilko H. Zwinderman, Springer, 2015.
- International Diabetes Federation (IDF) - <https://www.idf.org/>
- American Diabetes Association - <https://www.diabetes.org/>
- National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) <https://www.niddk.nih.gov/>